Acceptance gates (evaluated before any accuracy was inspected)

PASS GATE 1 gamma'=0 DMapS compile identical to 'none' (S_local, E_link, cost, circuit hash)
{'small': True, 'medium': True, 'large': True}

PASS GATE 2 small lambda calibrated on sigma^2 only: Bad<=0.80, Good>=0.05
G/M/B sigma2*M = 0.538/0.615/0.796 lambda=0.02117

PASS GATE 2 medium lambda calibrated on sigma^2 only: Bad<=0.80, Good>=0.05
G/M/B sigma2*M = 0.494/0.587/0.796 lambda=0.0143

PASS GATE 2 large lambda calibrated on sigma^2 only: Bad<=0.80, Good>=0.05
G/M/B sigma2*M = 0.402/0.530/0.798 lambda=0.01109

PASS GATE 3 A0_A3 mean sigma^2 == A3 mean sigma^2
max |diff| 2.2e-19

PASS GATE 4 same seed + same method -> bit-identical results

PASS GATE 5 CD-QFed vs base: identical channel uniforms and shot draws (CRN)
u a7a5e2e5b813/a7a5e2e5b813 z 7197db4bce3c/7197db4bce3c

PASS GATE 6 all sigma^2 in [0, 1/M]

PASS GATE 8 statevector backend self-test (torch/cuda kernels == numpy kernels)
backend=torch/cuda max|diff|=4.77e-07

PASS GATE 7 DMapS-FSNR selection never worse than baseline compile (predicted Psi_NSR)
{'small': ('fsnr10.0', {'none': 1.1261, 'fsnr0.1': 1.0043, 'fsnr1.0': 1.1198, 'fsnr10.0': 0.893}), 'medium': ('fsnr10.0', {'none': 1.1768, 'fsnr0.1':

PASS GATE 9 cross-device determinism (sharded multi-GPU sweep is sound)
max |diff| across devices = 0 | CRN hashes identical: True